SOFTWARE AND HARDWARE TOOLS

Task 1: Capture Network Traffic Using Wireshark

Aim: To capture network traffic for two minutes and identify the top three protocols.

Steps:

1. Download Wireshark from its official website: https://www.wireshark.org/ and install it.
2. Open Wireshark. Select the active Wi-Fi or Ethernet interface.
3. Click Start Capture, then browse Instagram or YouTube for about two minutes.
4. Click the red Stop button to stop capturing.
5. Save the capture using File > Save As and choose the .pcapng format (Wireshark's usual capture format). If your teacher specifically requires .pcap, select that format if available.
6. To see protocols, use Statistics > Protocol Hierarchy. Record the three protocols shown most often in your capture.

Task 2: Test Website Connectivity Using ping

Aim: To test connectivity to Spotify and Zomato and compare their response times.

Steps:

7. Open Command Prompt on Windows (or Terminal on Mac/Linux).
8. Run: ping www.spotify.com
9. After the command finishes, note the response times and the average if it is displayed.
10. Run: ping www.zomato.com
11. Compare the average response times. The smaller average is faster for this test. If a request times out, record it as timed out rather than making up a time.

Task 3: Check Running Processes and Memory Usage

Aim: To find processes related to Chrome, WhatsApp Desktop, or another browser app and record their memory usage.

Steps (Windows):

12. Press Ctrl + Shift + Esc to open Task Manager.
13. Open the Processes tab. Expand Google Chrome or WhatsApp if the entries are grouped.
14. Find the relevant processes and read the Memory column. Process names and memory use can change while apps are running.
15. Record the process name and memory usage. Do not close unfamiliar system processes just for this assignment.

Task 4: Scan Your Own Local Network

Aim: To identify at least two devices on a network that I own or have permission to scan.

Safety: Scan only your own home network or a college network when you have permission. Do not scan public or other people's networks.

Steps:

16. Download Angry IP Scanner from https://angryip.org/ or Advanced IP Scanner from its official website.
17. On Windows, you can first open Command Prompt and run ipconfig to find your IPv4 address and subnet mask.
18. Open the scanner and set the scan range to your own local subnet. Check the range carefully before starting.
19. Start the scan and wait for it to finish. Device names may be blank if a device does not provide one.
20. Record at least two devices that the scan actually finds. Do not guess device names or IP addresses.

Task 5: Three Hardware Tools for Network Troubleshooting

I used ChatGPT to identify three common hardware tools used for network troubleshooting.

Hardware tool	Purpose	When I would use it
Cable tester	Checks whether the wires in a network cable are connected correctly.	I would use it when a LAN cable does not work or may have a broken or incorrectly wired connection.
Crimping tool	Attaches an RJ45 connector to the end of an Ethernet cable.	I would use it when making or repairing a network cable.
Digital multimeter	Measures electrical values such as voltage, resistance, and continuity.	I would use it to check power supply or electrical continuity when troubleshooting compatible equipment, following safety precautions.

Final Conclusion

In this assignment, I learned how to capture network traffic using Wireshark, test website connectivity with ping, check running processes and memory usage, scan my own local network with permission, and identify hardware tools used for network troubleshooting. I will enter real results from my computer and attach my own screenshots.